GROVER SEARCH SIMULATION

DAY 26 | 30-DAY QUANTUM COMPUTING CHALLENGE | MINI PROJECT

1. Project objective

Build a small quantum search experiment over four possible outcomes: **00, 01, 10, 11**. Mark **10** as the target, amplify its probability using Grover's algorithm, and compare measurements before and after the search.

2. The quantum idea

A two-qubit system can represent four computational-basis states. Hadamard gates prepare an equal superposition. The oracle changes the **phase** of the marked state; the diffusion operator converts this phase difference into an increased measurement probability.

3. Circuit workflow

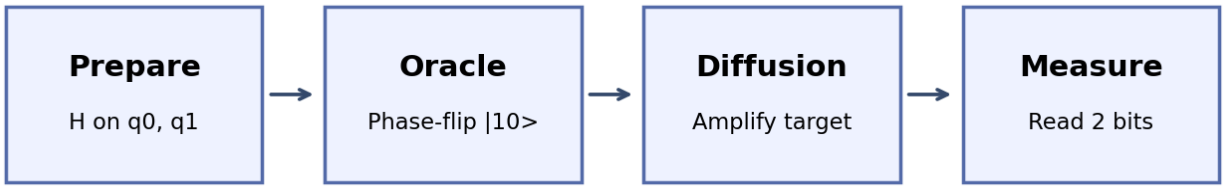


Figure 1. Conceptual Grover workflow. The circuit is simulated as matrices in the main notebook.

4. Essential mathematics

Initial state: |s> = (|00> + |01> + |10> + |11>) / 2.

Oracle: O|x> = -|x> when x is the target; otherwise O|x> = |x>.

Diffusion: D = 2|s><s| - I. One Grover iteration is G = DO.

For this four-state, one-target example, one iteration gives an ideal target probability of **100%**.

5. Why phase matters

The oracle alone does not increase the target measurement probability: it only flips its amplitude sign. Interference during diffusion is what makes the target more likely to be measured.

EXPERIMENT & EXECUTION

DAY 26 | PRACTICAL WALKTHROUGH

6. Step-by-step experiment

Step 1. Install Python packages: `pip install numpy matplotlib notebook`.

Step 2. Open `grover_search_simulation.ipynb` in Jupyter or VS Code.

Step 3. Execute the main experiment cell. It prepares the statevector, builds the oracle and diffusion matrices, and samples 1,000 measurements.

Step 4. Compare the before/after histograms. Optionally install Qiskit Aer and run the hardware-style circuit simulator cell.

7. Worked amplitude example

Before the oracle: `[+0.5, +0.5, +0.5, +0.5]`.

After marking 10: `[+0.5, +0.5, -0.5, +0.5]`.

The mean amplitude is 0.25. Diffusion reflects each amplitude about the mean: a becomes 2(0.25) - a.

After diffusion: `[0, 0, 1, 0]`. Thus, measuring the ideal final state returns **10** with certainty.

8. Expected outcomes

State	Before: theory	After: theory
00	25%	0%
01	25%	0%
10	25%	100%
11	25%	0%

Measured frequencies before Grover fluctuate around 25%; the post-Grover ideal distribution is deterministic for this example.

RESULTS & INTERPRETATION

DAY 26 | RESULTS, LIMITATIONS AND SUMMARY

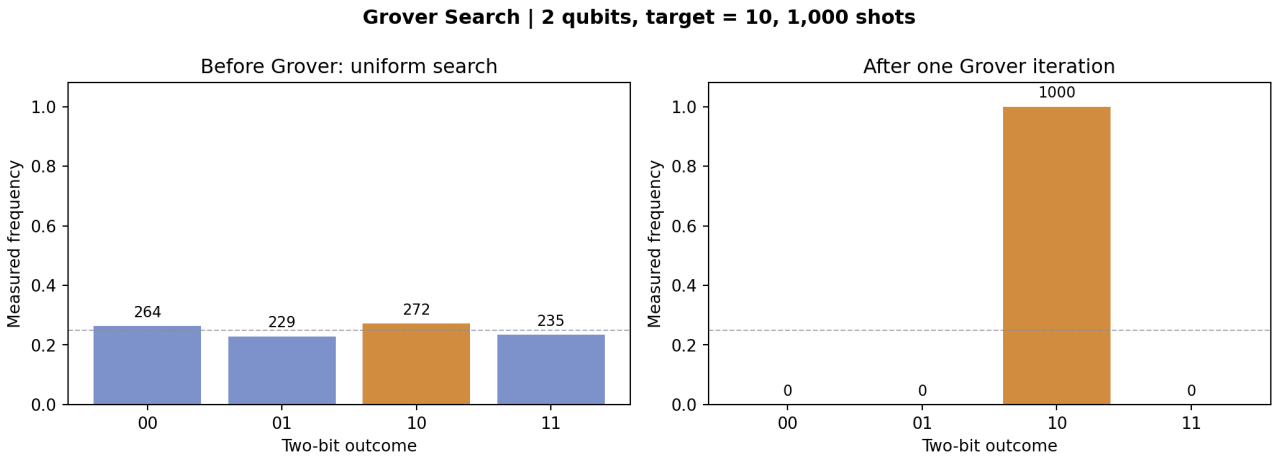


Figure 2. Reproducible seeded results, 1,000 shots, target 10.

9. What the results mean

The initial distribution is spread across all four outcomes. After one ideal Grover iteration, the probability mass is concentrated on the marked state. This is **amplitude amplification**, not a random lucky guess.

10. Limitations and scientific accuracy

This experiment uses a **noiseless classical statevector simulation**. It does not demonstrate speedup over a classical search on real hardware. The exact 100% success after one iteration is a special property of a four-item search with one marked item; larger search spaces require different iteration counts.

11. Try it yourself

Change the marked target; vary the number of shots; remove the diffusion step; compare the oracle-only probabilities; optionally run the equivalent Qiskit Aer circuit.

Summary: What to Remember

Superposition prepares candidate states. **Oracle** marks the target by phase. **Diffusion** amplifies its amplitude through interference. **Measurement** reveals the target with high probability after an appropriate number of iterations.

GitHub files: Grover_Search_Simulation.pdf, grover_search_simulation.ipynb, results.png, circuit.png, README.md and requirements.txt.